

Differential Effects of Virtual and Augmented Reality on Social Presence and Engagement in Collaborative Gaming for Unfamiliar Users

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ABSTRACT

Many studies have shown that collaborative tasks in immersive virtual reality (VR) and augmented reality (AR) environments can enhance collaborative outcomes in learning, training, and game-based contexts. However, the literature offers limited information on how these environments differentially shape social presence and collaborative engagement, particularly among unfamiliar users. This study addresses this gap by examining differences in social presence and collaborative engagement between VR and AR environments for unfamiliar pairs. A between-subjects experiment used an escape room game that featured three collaborative tasks, identically implemented in both VR and AR. The key difference was that the VR experience was stationary, while the AR experience required physical movement between rooms. The study involved 52 participants, divided into VR and AR groups, where two unfamiliar participants were paired into teams to complete the tasks. The results indicate significant differences in collaborative dynamics and user perception between the two environments. Specifically, VR pairs reported a stronger sense of immersion and flow state, whereas AR pairs demonstrated greater contextual awareness and behavioral coordination. Cybersickness measures also differed between conditions; given the locomotion mismatch, this pattern should be interpreted cautiously and not attributed to the environment alone. This finding improves understanding of the impact of immersive environments on collaborative processes and offers insights for designing collaborative XR applications (e.g., training and game-based teamwork), particularly for unfamiliar users.

Index Terms: Collaborative Interaction, Virtual Reality, Augmented Reality, Interaction Design, Cybersickness.

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1 INTRODUCTION

Collaborating with strangers occurs frequently in project teams and professional training, but it poses significant challenges. Unfamiliar members must build shared understanding, coordinate roles, and maintain engagement without established trust or relationships. These conditions often amplify communication breakdowns and attentional lapses, making teamwork cognitively demanding and socially complex [12, 24]. Institutions have increasingly adopted extended reality (XR), which includes virtual reality (VR) and augmented reality (AR), to support co-location collaboration in ad-hoc projects [6], training scenarios [8], and collaborative gamified educations. Existing studies demonstrate that VR and AR improve social interaction and participation [41], indicating a strong potential for immersive collaboration. However, collaboration remains especially difficult when participants lack prior familiarity, as they must quickly establish common ground (a shared understanding of knowledge and assumptions relevant to the task [14]), coordinate actions and maintain engagement under comfort and attention constraints [12].

Most studies report the potential of XR to increase interaction and cohesion, but direct comparisons between VR and AR focus mainly on individual experiences or familiar teams [40]. These approaches overlook scenarios in which unfamiliar individuals must collaborate. In randomly assigned teams, members typically lack a shared interaction history, which intensifies the challenge of quickly establishing common ground and effective collaboration. Therefore, conducting a comparative analysis is essential to assess the advantages and disadvantages of AR and VR in collaborative scenarios [53]. In the absence of this evidence, XR designs may overlook the needs of ad-hoc teams, potentially reducing effectiveness in collaboration-sensitive environments [47].

VR and AR provide contrasting balances between immersion and real-world awareness. VR immerses users within a shared virtual environment, promoting deep absorption but potentially increasing visual-vestibular conflicts [50]. AR overlays virtual content on physical surroundings, supporting face-to-face cues that improve mutual attention and situated coordination while splitting focus between real and virtual elements [51, 10]. These differences in modality influence social presence, collaborative engagement, and cybersickness in different ways [54, 30, 57]. However, few studies examine these factors in the same co-located task involving unfamiliar participants. Earlier studies [17] compared VR and AR, but focused on single-user contexts, leaving collaborative multi-user settings unexplored. Collaborative puzzle games provide a con-

trolled yet ecologically relevant testbed for studying ad hoc teamwork, as they require rapid coordination and sustained engagement from all participants to solve puzzles.

This study aims to explore the distinct impacts of VR and AR conditions on unfamiliar users during a collaborative puzzle game, to identify the strengths and weaknesses of each modality to guide effective design approaches for collaborative XR applications. Specifically, the study focuses on addressing the following research questions (RQs):

RQ1: How do levels of social presence and task engagement differ between AR and VR environments for unfamiliar users performing collaborative gaming tasks?

RQ2: What are the perceived challenges to social interaction for unfamiliar users during collaborative gaming in AR and VR?

The primary contributions include:

- Perform a thorough analysis of the interaction between unfamiliar users in collaborative VR and AR gaming tasks, concentrating on how each modality influences social presence and collaborative participation.
- Conduct a comprehensive mixed-method study to identify key challenges and limitations that impact social interactions among unfamiliar users engaged in collaborative gaming tasks within VR and AR environments.

2 RELATED WORK

Existing studies on immersive collaboration have investigated how AR and VR shape social presence, engagement, and communication processes. However, existing findings are often fragmented in modality comparisons, social presence research, and task-based collaboration studies, making it difficult to directly infer how immersive amenities influence collaborative interaction. The related work is organized into two subsections: (1) modality effects on social presence and engagement (RQ1), and (2) perceived challenges to social interaction and coordination in AR and VR (RQ2).

2.1 Modality Effects on Social Presence and Task Engagement

Many studies had demonstrated that immersive modality systematically affects presence, engagement, and experiential quality, providing theoretical foundation for comparing AR and VR collaboration. Verhulst *et al.* [51] compare the implementation of VR and AR of a cultural experience and report modality-dependent differences in presence and engagement, indicating that immersive media are not experientially equivalent. Greenfeld *et al.* [17] similarly identify modality-specific variations in emotional response and engagement across immersive platforms. Voit *et al.* [52] further show that the evaluation context, including AR, VR, laboratory, and in-situ settings, significantly influences outcome measures, highlighting the importance of controlled modality comparisons when examining collaborative effects.

Within collaborative contexts, AR is frequently associated with increased situational awareness due to visibility in-situ and shared physical space. Billinghurst *et al.* [4] argue that AR collaboration benefits from real-world grounding that supports mutual awareness and deictic referencing, suggesting potential advantages for social presence in co-located tasks. Gottsacker *et al.* [15] extend this view by showing that symmetry of immersion in co-located AR settings modulates engagement and social interaction, indicating that AR-related presence is shaped not only by visibility but also by how immersion is distributed among collaborators. Zhou *et al.* [59] further demonstrate that AR and VR elicit different attention allocation and coordination strategies, implying that task engagement develops through modality-specific interaction patterns.

In contrast, VR collaboration has been shown to promote focused immersion and sustained engagement. Dow *et al.* [13] report that immersive virtual environments support deep engagement and narrative flow. Han *et al.* [18] argue that VR's environmental shielding and controllability facilitate coordinated group action, although at the cost of removing real-world social cues. Oh *et al.* [35] identify avatar-based representation, such as gaze, gesture, and expressive channels, as essential to maintain social presence in VR.

Empirical studies further demonstrate that richer avatar expressivity and realism increase social presence and interpersonal engagement. Wu *et al.* [56] show that a higher avatar expressivity leads to a greater social presence and improved interpersonal outcomes in pair collaboration. Freeman *et al.* demonstrate that avatar-mediated self-presentation shapes how users perceive themselves and others, influencing interaction dynamics. Yousefdeh *et al.* [58] report that avatar realism and communication richness strengthen co-presence and collaboration quality. Li *et al.* [29] further show that stylized gaze cues increase joint attention and reduce verbal overhead. Together, these findings establish that AR and VR support social presence and engagement through different cost profiles, motivating direct comparison with identical collaborative tasks.

2.2 Perceived Challenges to Social Interaction and Coordination

In addition to general presence and engagement, existing work highlights modality-specific challenges that affect how users perceive and manage social interaction during collaboration. In AR, perceptual and technical constraints can undermine coordination despite a strong real-world foundation. Kruijff *et al.* [26] identify divided attention, occlusion, and unstable registration as sources of increased coordination cost. Aslan *et al.* [2] show that combining AR with other media fragments attention and reshapes interaction behavior, suggesting that AR's openness to the physical environment can complicate sustained collaborative focus. These findings indicate that, while AR supports the visibility of teammates, it may introduce interactional challenges that affect the perceived communication effort.

In VR, challenges arise from the absence of physical co-presence and reliance on mediated cues. Lee *et al.* [28] demonstrate that avatar fidelity and available communication channels influence perceived agency, which in turn shapes interaction quality. Sayadi *et al.* [40] report that while social VR research emphasizes gesture and embodiment features, users often struggle to articulate how these features affect collaboration, pointing to challenges in interpreting and utilizing platform features. MacArthur [31] further argues that benefits are socially constructed during interaction, meaning that perceived challenges originate not only from system limitations but also from how users interpret and adapt to available cues.

Studies involving unfamiliar collaborators further highlight these challenges. Pan *et al.* [37] show that embodiment cues help strangers coordinate but do not eliminate initial uncertainty. Agnew *et al.* [1] demonstrate that stylized avatars can promote rapport among unfamiliar teammates, but do not fully resolve coordination breakdowns. In applied collaborative scenarios, Ioannou *et al.* [20] and Pazakou *et al.* [38] show that multi-user VR escape rooms elicit engagement but also surface communication bottlenecks during joint problem-solving.

The task structure has been shown to amplify these perceived challenges. Karaosmanoglu *et al.* [21] and Grandi *et al.* [16] demonstrate that asymmetric tasks increase interdependence and communication demands, making coordination difficulties more evident. These findings motivate the use of asymmetric collaborative gaming tasks to surface modality-specific challenges to social interaction.

2.3 Summary and Positioning

Early comparative work by Tang *et al.* [46] identifies differences in the presence between AR and VR during social interaction, while subsequent studies clarify how engagement, coordination, and communication challenges arise from modality-specific features. However, few studies systematically examine how unfamiliar users experience social presence, engagement, and interaction challenges across AR and VR within the same co-located collaborative task. Through grounding the experimental design in previous findings on modality affordances, non-verbal communication, and task asymmetry, this study addresses the gap and empirically examines differences in social presence and task engagement between AR and VR collaborative gaming environments, along with the perceived challenges to social interaction among unfamiliar users.

3 METHODOLOGY

This study uses a collaborative two-player dungeon escape game (see Fig. 1) consisting of three mini-games, each in its own room. To ensure environmental consistency in AR and VR conditions, the dimensions of the virtual room match exactly the physical testing space. Each mini-game uses one of three asymmetry mechanisms: informational, positional, or visual, to structure interdependent collaboration. The game design is based on the foundational work [22] and applies the Octalysis framework [9]. Both VR and AR implementations share identical core game mechanisms, including task objectives, rule sets, and data log, while ensuring that virtual layouts, object dimensions, and visual assets remain consistently aligned in both settings. In AR condition, participants viewed the physical room via headset passthrough as visual baseline, with a semi-transparent surface mesh overlaid on surfaces to provide spatial mapping, boundary awareness, and stable alignment of virtual content with the physical environment. Importantly, participants retained direct visibility of the real world and their teammate's body and actions.

The key difference between settings is the teammate's representation: in AR, players see both their teammate's real body and a virtual avatar overlay, whereas in VR they see only the teammate's virtual embodiment. In both conditions, the avatars comprised a head and two hands; the head was driven by HMD tracking, the hands by tracked controllers (supporting gesture cues), and the gaze direction was inferred from the head orientation. The other difference is that players in the VR setting require minimal physical movement within a safe, confined space, as virtual movement is controlled via a controller, whereas the AR setting requires players to physically navigate the room. Controller-based locomotion in VR was selected because the experimental space was relatively small and contained physical obstacles; enabling free walking for two co-located VR users wearing opaque HMDs would have posed safety risks (e.g., collisions or tripping). The game was implemented in Unity and was deployed on two head-mounted Meta Quest 3 displays; the source code is publicly available [11].

3.1 Mini-Game 1: Animal Identification Puzzle

The first mini-game uses informational asymmetry within its task design [3], as illustrated in Figs. 2a, 2b, 2c, and 2d. Two players attempt to solve their respective puzzles based on the clues provided by rotating a wheel to select the correct animal. Player A has access to the three clues required by player B (and vice versa), where these clues are not visible to each other. This asymmetric task design mandates verbal information exchange between players, which also provides an opportunity for both players to collaboratively discuss the clues. A shared health bar, comprising three hearts, loses one heart per incorrect selection; the game ends when all hearts are depleted, thus introducing collective risk and encouraging discussion before making a decision. A hint feature is provided to mitigate frustration and maintain engagement [32].

3.2 Mini-Game 2: Potion Brewing Puzzle

The second mini-game adopts a co-located positional asymmetry [55] within its task design by positioning players in separate locations, as illustrated in Figs. 2e, 2f, 2g, and 2h. Only Player A can activate the floor trigger to unlock the chamber door, enabling Player B to enter the chamber and view three linear-equation problems hidden behind a bookcase. These equations remain invisible to player A in the AR setting. If player A moves away from the trigger, the chamber door locks, preventing simultaneous access. Player B then solves the equations or collaboratively discusses the problems and solutions with Player A. These solutions provide the formula player A needs to brew the correct potion, thus completing the task.

3.3 Mini-Game 3: Shape Alignment Puzzle

The third mini-game features a visual asymmetry [36] in which each player has a unique perspective, as illustrated in Figs. 2i, 2j, 2k, and 2l. Only player A can see the pattern on the leftmost cube, whereas only player B can see the pattern on the rightmost cube. The players must manipulate the four central cubes to form a composite drawing by aligning all patterns correctly to complete this task. Similarly, this task requires both players to verbally describe their unique patterns to each other, promoting social interaction.

4 EXPERIMENT DESIGN

4.1 Participants

The study used the between-subjects approach and recruited 52 university students (aged 18 to 26 years; $\mu_{age} = 21$, $\sigma_{age} = 1.8$) through campus social media channels. All participants indicated having no previous experience with VR or AR technologies and were randomly paired with an unfamiliar teammate, resulting in 26 teams, each assigned to AR or VR condition. Table 1 presents the demographics of the participants.

4.2 Procedure

Figure 3 presents an overview of the experimental procedure. The session began with an introduction to the study, after which the participants provided their informed consent. A 15-minute tutorial was conducted to familiarize participants with the game operations. Each team then completed collaborative game design tasks that took an average of 20 to 30 minutes (the maximum recorded completion time was 40 minutes). Participants maintained the right to withdraw from the study at any time if they felt discomfort. Following a 5-minute rest period, participants individually completed self-rating questionnaires. Then a one-on-one interview was conducted, lasting approximately 20 minutes, to collect qualitative feedback from the participants. The entire procedure lasted an average of 60 to 70 minutes. The study protocol was reviewed and approved by the Code of Practice for Research Ethics Committees of the University of Nottingham Ningbo China [49].

4.3 Measurements

This study uses a mixed-methods approach to collect quantitative and qualitative data. Quantitative data was obtained from two benchmark questionnaires: the Game Experience Questionnaire (GEQ) [19] and the Simulator Sickness Questionnaire (SSQ) [23]. Qualitative data consist of feedback from semi-structured interviews that provides in-depth participant insight.

The GEQ provides a modular framework that is well-suited to address both RQs. The *Core* (CM) module quantifies immediate gameplay experiences in seven components. The *immersion*, *flow* and *positive affect* scores serve as metrics for collaborative engagement (RQ1), while *tension*, *negative affect*, and *challenge* help identify experiential friction points (RQ2). The *Social Presence* (SP) module, administered due to the co-located collaborative nature of the activity, offers a standardized measure of psychological

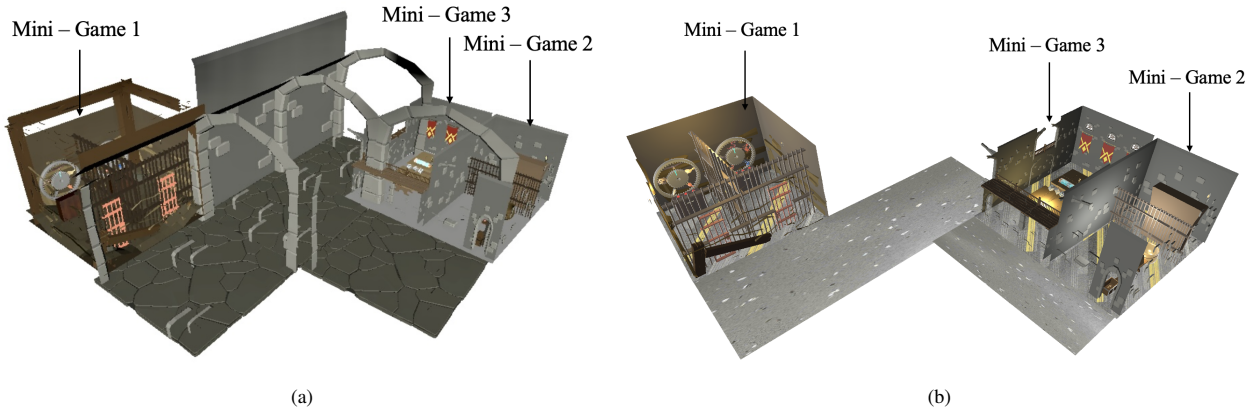


Figure 1: Environmental layouts for the VR (a) and AR (b) conditions. Each setup consists of three separate rooms, one per mini-game, with virtual room dimensions matched to the physical space to ensure consistency across environments.

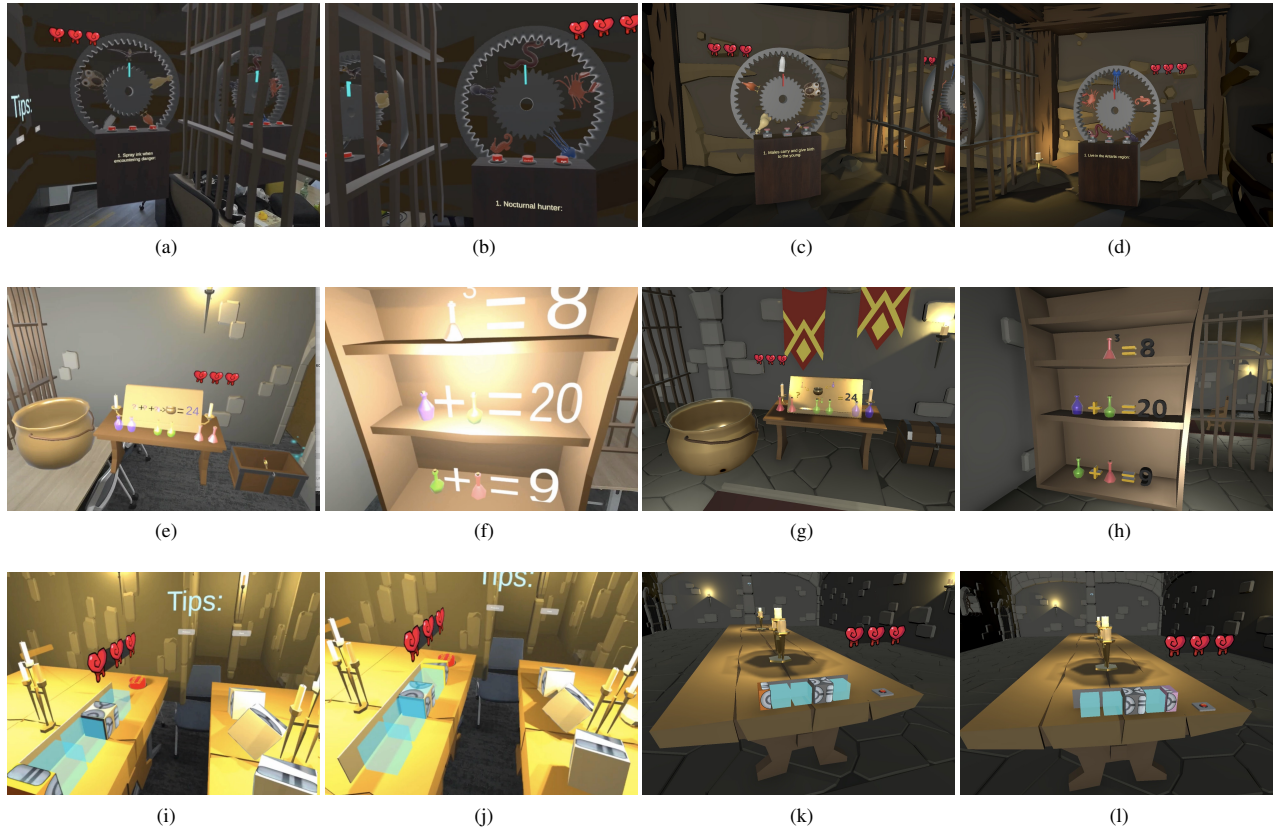


Figure 2: Player perspectives within both AR and VR conditions for each mini-game. Row 1 (mini-game 1): (a) Player A AR, (b) Player B AR, (c) Player A VR, (d) Player B VR; Row 2 (mini-game 2): (e) Player A AR, (f) Player B AR, (g) Player A VR, (h) Player B VR; Row 3 (mini-game 3): (i) Player A AR, (j) Player B AR, (k) Player A VR, (l) Player B VR.

and behavioral involvement with another player, allowing a structured comparison between the AR and VR conditions. This operationalization aligns with the focus of RQ1 on how unfamiliar partners establish common ground and maintain engagement during joint action. The *Post-Game* (PG) module captures reflective

assessments, allowing participants to articulate specific challenges and limitations that affected their social interactions after reflecting on the entire experience (RQ2). This modular structure enables the GEQ to assess in-the-moment sensations, the quality of social interactions, and post-hoc reflections relevant to the social dynamics



Figure 3: Experimental procedure that consists of three phases: a 20-minute pre-game phase (introduction, consent, tutorial), a 20- to 30-minute collaborative gameplay phase, and a 25-minute post-game phase (rest, questionnaires, interview).

Table 1: Demographic data of the 52 participants assigned into pairs and subjected to AR or VR conditions. “PID” indicates the pair number, and “A” and “B” designate players A and B within each pair.

VR				AR			
PID	Faculty	Age	Gender	PID	Faculty	Age	Gender
1A	FoSE	20	Male	3A	FoSE	19	Male
1B	FoSE	20	Male	3B	FoSE	20	Female
2A	FoSE	20	Female	13A	FoSE	20	Female
2B	FoSE	21	Female	13B	NUBS	22	Female
4A	NUBS	20	Female	16A	NUBS	21	Male
4B	FHSS	19	Female	16B	FHSS	22	Male
5A	FoSE	20	Female	17A	FHSS	23	Male
5B	FoSE	20	Female	17B	NUBS	21	Male
6A	NUBS	20	Male	18A	FoSE	20	Male
6B	FoSE	20	Female	18B	NUBS	21	Female
7A	NUBS	20	Male	19A	FHSS	22	Female
7B	FHSS	21	Male	19B	FHSS	22	Female
8A	FoSE	22	Female	20A	FHSS	23	Female
8B	FHSS	20	Male	20B	NUBS	21	Male
9A	CELE	18	Male	21A	FoSE	20	Female
9B	CELE	18	Male	21B	NUBS	21	Male
10A	FoSE	22	Female	22A	CELE	19	Female
10B	NUBS	23	Male	22B	NUBS	21	Female
11A	FHSS	23	Male	23A	FoSE	20	Female
11B	NUBS	25	Male	23B	FoSE	20	Female
12A	NUBS	26	Female	24A	FoSE	19	Male
12B	FHSS	26	Female	24B	FoSE	21	Male
14A	NUBS	24	Female	25A	NUBS	22	Female
14B	FoSE	22	Female	25B	NUBS	21	Female
15A	CELE	19	Female	26A	FHSS	23	Female
15B	CELE	18	Male	26B	FHSS	22	Female

CELE: Centre for English Language Education
FoSE: Faculty of Science and Engineering
NUBS: Nottingham University Business School
FHSS: Faculty of Humanities and Social Sciences

of collaborative games within AR and VR conditions.

The SSQ provides an important measure of the adverse physio-

logical side effects resulting from exposure to immersive technologies. SSQ quantifies the severity of symptoms on three different subscales, including *nausea*, *oculomotor disturbances*, and *dizziness*. These data directly identify physical challenges (RQ2), such as vertigo, headache, or nausea, that can severely impact a player’s capacity for social interaction and collaboration within AR and VR conditions. Differences between-conditions in SSQ scores provide a quantitative index of physical discomfort that can be associated with diminished social presence or collaborative participation, although such links should be interpreted carefully given potential confounding factors (e.g. locomotion).

The semi-structured interview was designed to collect qualitative data on user perceptions of collaborative engagement and social presence (RQ1), as well as specific challenges encountered (RQ2). The questions prompted participants to reflect on the factors that both facilitated and affected their collaborative experience within the AR or VR environment. Table 2 presents the open-ended interview questions used in this study.

5 RESULTS

5.1 GEQ Results

Table 3 summarizes the GEQ scores, revealing significant differences in user experience between VR and AR conditions. The non-normal distributions, validated by Shapiro-Wilk tests ($W(p)$), led to the use of Mann-Whitney U tests for analysis. This section presents a item-level analysis that groups participant ratings by component within each module and references items using a “module-item” format with full item descriptions are provided in [27].

5.1.1 Core Module

Participants in the AR condition showed a significantly higher *competence* score than participants in the VR condition ($U = 462.5, p = 0.021^*, d = 0.719$). In contrast, participants in VR condition produced significantly higher *sensory and imaginative immersion* ($U = 195.5, p = 0.009^{**}, d = 0.701$), a higher *flow state* ($U = 54.5, p < 0.001^{***}, d = 2.053$) and an elevated perceived *challenge level* ($U = 162.5, p = 0.001^{***}, d = 1.058$). No significant differences were found for *tension*, *negative affect*, or *positive affect*. The item-level analysis (see Fig. 4) also revealed that partici-

Table 2: Semi-structured interview questions.

No.	Question	Dimension
1	To begin, can you walk me through your experience from start to finish?	General Impression
2	What stands out to you most when you reflect on the game activity?	
3	Can you describe any physical sensations you noticed while playing the game?	Physical Sensations
4	What was it like to collaborate with your teammate?	Collaboration Process
5	What, if any, features of the [AR/VR] environment facilitated your collaboration, and what impeded successful teamwork?	
6	How did the [AR/VR] environment affect your collaboration with your teammate?	Technology Role

Table 3: GEQ scores results, presenting the mean (μ), median (M), standard deviation (σ), interquartile ranges (IQR), test statistic (U), p and Cohen's d for each component. Shapiro–Wilk tests ($W(p)$) confirmed non-normal distributions, leading to Mann-Whitney U tests between VR and AR conditions in three modules: *Core*, *CR* (competence (COMP), sensory and imaginative immersion (SENS), flow (FLOW), tension (TENS), challenge (CHAL), negative affect (NEGA), positive affect (POSA)), *Social Presence*, *SP* (psychological involvement - empathy (PSYI-E), psychological involvement - negative feelings (PSYI-N), behavioral involvement (BEHI)), and *Post-Game*, *PG* (positive experience (POSE), negative experience (NEGE), tiredness (TIRD), returning to reality (RETR)).

Module	Component	VR					AR					p	Cohen's d	U
		μ	M	σ	IQR	$W(p)$	μ	M	σ	IQR	$W(p)$			
CR	COMP	2.962	3.000	0.727	0.750	0.901 (0.016*)	3.381	3.400	0.372	0.400	0.874 (0.004**)	0.021**	0.719	462.5
	SENS	3.340	3.500	0.536	0.333	0.729 (< 0.001***)	2.891	3.000	0.729	0.917	0.909 (0.025*)	0.009**	0.701	195.5
	FLOW	3.046	3.300	0.727	1.200	0.858 (0.002**)	1.754	1.800	0.513	0.400	0.894 (0.011*)	< 0.001***	2.053	54.5
	TENS	1.808	1.000	1.067	2.000	0.736 (< 0.001***)	1.423	1.333	0.486	0.917	0.791 (< 0.001***)	0.690	0.464	317.5
	CHAL	2.123	2.000	1.254	2.150	0.927 (0.046*)	1.023	0.800	0.768	1.050	0.858 (0.002**)	< 0.001***	1.058	162.5
	NEGA	0.942	1.000	0.356	0.500	0.918 (0.040*)	0.904	0.750	0.553	0.750	0.862 (0.002**)	0.391	0.083	292
SP	POSA	3.362	3.400	0.509	0.800	0.926 (0.049*)	3.354	3.400	0.464	0.750	0.933 (0.049*)	0.912	0.016	332
	PSYI-E	3.532	3.500	0.346	0.500	0.920 (0.045*)	3.673	3.667	0.260	0.292	0.851 (0.002**)	0.127	0.460	419.5
	PSYI-N	1.723	1.800	0.371	0.600	0.918 (0.041*)	2.100	2.000	0.279	0.200	0.854 (0.002**)	< 0.001***	1.148	538.5
	BEHI	3.526	3.500	0.193	0.167	0.894 (0.012*)	3.660	3.833	0.314	0.292	0.820 (< 0.001***)	0.015*	0.516	477.5
	POSE	3.577	3.667	0.246	0.417	0.853 (0.002**)	3.417	3.500	0.306	0.333	0.820 (< 0.001***)	0.030**	0.577	222
	NEGE	0.564	0.083	0.741	1.333	0.727 (< 0.001***)	0.276	0.167	0.333	0.458	0.785 (< 0.001***)	0.803	0.502	325
PG	TIRD	1.154	0.500	1.287	2.500	0.804 (< 0.001***)	0.712	0.500	0.874	1.000	0.775 (< 0.001***)	0.304	0.402	284
	RETR	2.808	2.667	0.737	1.333	0.842 (< 0.001***)	2.115	2.000	0.766	0.583	0.802 (< 0.001***)	< 0.001***	0.921	138.5

* $p < .05$; ** $p < .01$; *** $p < .001$

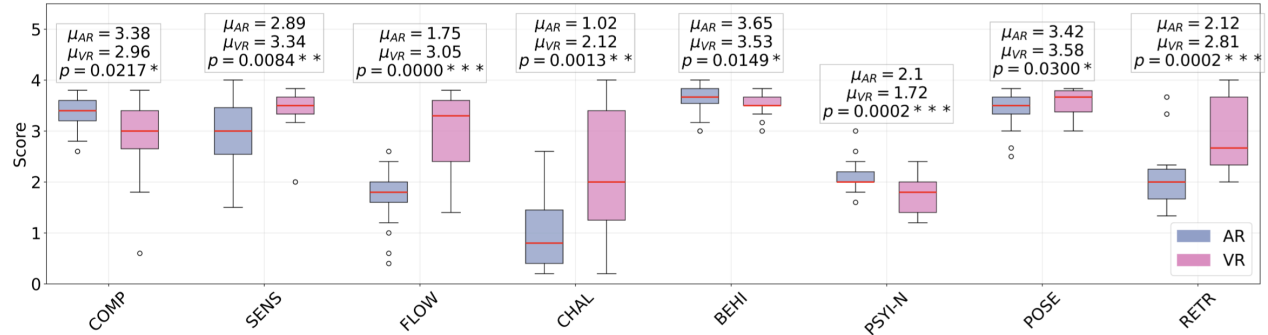


Figure 4: Item-level analysis of GEQ scores between AR and VR conditions, showing components with statistically significant differences. The mean values (μ) and p are reported for each component: competence (COMP), sensory and imaginative immersion (SENS), flow (FLOW), challenge (CHAL), psychological involvement - negative feelings (PSYI-N), behavioral involvement (BEHI), positive experience (POSE) and return to reality (RETR).

pants in the AR condition reported feeling more competent (CR-10, $p = 0.011^*$) and faster in achieving the objectives (CR-21, $p < 0.001^{***}$). In contrast, participants in VR condition indicated stronger imaginative sensations (CR-18, $p = 0.002^{**}$) and a greater sense of exploration (CR-19, $p = 0.006^{**}$). All flow-related components (CR-5, CR-13, CR-25, CR-28, CR-31) scored significantly higher for participants in VR condition ($p < .001$). Furthermore,

participants in the VR condition reported a greater perceived difficulty (CR-11, $p = 0.001^{***}$), pressure (CR-23, $p = 0.049^*$), challenge (CR-26, $p = 0.005^{**}$), time pressure (CR-32, $p = 0.046^*$) and effort (CR-33, $p = 0.005^{**}$).

Table 4: SSQ scores results, presenting the mean (μ), median (M), standard deviation (σ), interquartile range (IQR), test statistic (U), p and Cohen's d for each subscale. Shapiro–Wilk tests ($W(p)$) confirmed non-normal distributions, leading to Mann–Whitney U tests between VR and AR conditions in three subscales: *nausea*, *oculomotor*, and *disorientation*.

Subscale	VR					AR					p	Cohen's d	U
	μ	M	σ	IQR	$W(p)$	μ	M	σ	IQR	$W(p)$			
Nausea	53.938	38.160	42.230	81.090	0.836 ($< 0.001^{***}$)	20.548	9.540	35.562	26.235	0.622 ($< 0.001^{***}$)	$< 0.001^{***}$	0.855	130.5
Oculomotor	38.192	30.320	25.501	22.740	0.840 ($< 0.001^{***}$)	29.737	22.740	26.942	15.160	0.735 ($< 0.001^{***}$)	0.049*	0.322	233.0
Disorientation	69.600	48.720	56.509	97.440	0.874 (0.004**)	92.622	69.6	72.056	62.640	0.821 ($< 0.001^{***}$)	0.165	0.356	413.5
Total	60.438	59.840	29.777	41.140	0.855 ($< 0.001^{***}$)	47.613	29.92	44.924	20.570	0.702 ($< 0.001^{***}$)	0.044*	0.28	228

* $p < .05$; ** $p < .01$; *** $p < .001$

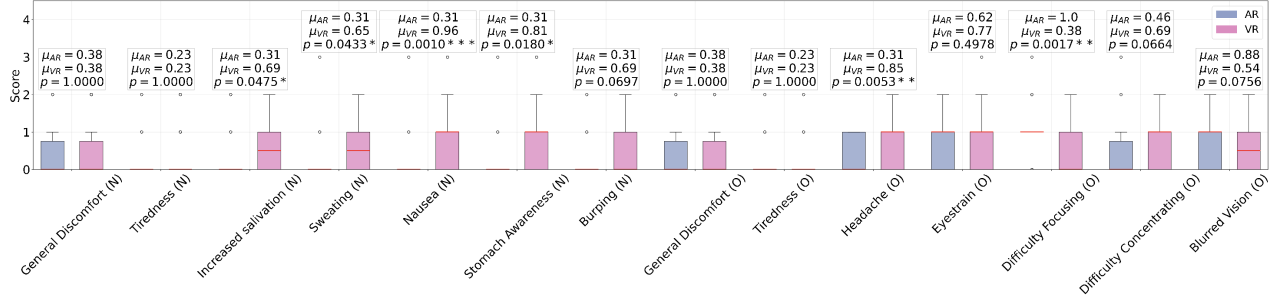


Figure 5: Symptom-level analysis of SSQ scores between AR and VR conditions, showing items with statistically significant differences. The mean values (μ) and p are reported for each symptom: *nausea* (N) (general discomfort, tiredness, salivation, sweating, nausea, stomach awareness) and *oculomotor discomfort* (O) (general discomfort, tiredness, headache, difficulty focusing).

5.1.2 Social Presence Module

Participants in the AR condition showed a significantly higher score for *psychological involvement - negative feelings* ($U = 538.5, p < 0.001^{***}, d = 1.148$) and *behavioral involvement* ($U = 477.5, p = 0.015^*, d = 0.516$), but did not show a significant difference in *psychological involvement - empathy*. The item-level analysis indicated that participants in the AR condition felt more influenced by the moods of others (SP-12, $p < 0.001^{***}$) and reported a slightly greater influence on the moods of others (SP-11, $p = 0.040^*$). For *behavioral involvement*, participants in the AR condition reported greater mutual attention (SP-5, $p < 0.001^{***}$; SP-6, $p = 0.013^{**}$).

5.1.3 Post-Game Module

Participants in VR condition showed a significantly higher *positive experience* ($U = 222, p = 0.029^*, d = 0.577$) and a significantly higher *return to reality* ($U = 138.5, p < 0.001^{***}, d = 0.921$). No significant differences were found for *negative experience* or *tiredness*. The item-level analysis revealed that participants in VR condition felt more revived (PG-1, $p = 0.001^{***}$), found it more difficult to return to reality (PG-3, $p = 0.007^{**}$) and experienced a stronger sense of returning from the game (PG-17, $p < 0.001^{***}$).

5.2 SSQ Results

Table 4 presents the SSQ scores and highlights significant between-condition differences in physical discomfort, with Shapiro–Wilk tests confirming non-normal distributions, which justify the use of Mann–Whitney U tests. Participants in VR condition showed markedly higher *nausea* than participants in AR condition ($U = 130.5, p < 0.001^{***}$, Cohen's $d = 0.855$) and modestly higher *oculomotor discomfort* ($U = 233, p = 0.049^*, d = 0.322$). No significant differences in *disorientation* were found despite a higher median score for participants with AR condition ($U = 413.5, p = 0.165$). The overall median SSQ score was also significantly higher for participants in VR condition ($U = 228, p = 0.044^*, d = 0.28$). The symptom-level analysis (see Fig. 5) corroborated these ef-

fects. Compared to the participants in the AR condition, for participants in the VR condition showed increased *general discomfort* ($p < 0.001^{***}$), *tiredness* ($p = 0.006^{**}$), *increased salivation* ($p = 0.046^*$), *sweating* ($p = 0.042^*$), *nausea* ($p = 0.001^{**}$), *stomach awareness* ($p = 0.017^*$), as well as oculomotor-related *headache* ($p = 0.005^{**}$) and *difficulty focusing* ($p = 0.002^{**}$). Given the unmatched locomotion between conditions, these SSQ differences should be interpreted as condition-level effects rather than effects attributable solely to the environment.

5.3 Interview Results

Table 5 presents the thematic analysis of the interview responses, which reveals distinct differences in social engagement between participants under AR and VR conditions. The analysis indicates clear differences in the way each condition promoted social presence and collaborative engagement. In VR, 88.46% of the responses highlighted positive teamwork, task-focused coordination, and mutual trust, suggesting that the fully VR environment supported a strong sense of shared purpose. However, the same environment showed a certain restriction in non-verbal communication, with 11.54% of the responses mentioning communication lags. In contrast, in AR, 73.08% of the responses emphasized the benefits of face-to-face interaction and the availability of non-verbal communication, which improved interpersonal awareness and coordination.

The participants identified different challenges that impact social interactions in each condition. In VR, 30.77% of the responses reported physical discomfort such as motion sickness and eye strain, which could detract from social engagement. Furthermore, 23.08% recognized the absence of subtle non-verbal communication in VR settings as a limitation. In AR, 15.38% of the participants identified environmental distractions as a key challenge affecting shared attention. Furthermore, 26.92% of AR responses reported increased self-consciousness or initial shyness, indicating that increased social visibility could also intensify social sensitivity.

Table 5: Thematic analysis of interview responses on the gaming experiences of 52 participants in AR or VR conditions.

Dimension	Experience	VR			AR		
		Component	Count	Total (%)	Component	Count	Total (%)
General Impression	Positive	Immersion Achievement Novelty	9 8 6	23 (88.46%)	Interesting Novelty Social Enjoyment	10 7 5	22 (84.62%)
	Negative	Loss of Real-world Awareness	3	3 (11.54%)	Distraction During Gameplay	4	4 (15.38%)
Physical Sensations	Positive	Full-body Immersion Flow Experience	10 8	18 (69.23%)	Natural Movement Spatial Awareness	12 9	21 (80.77%)
	Negative	Motion Sickness Eye Strain	6 2	8 (30.77%)	Motion Sickness Visual Misalignment	2 3	5 (19.23%)
Collaboration Process	Positive	Task-focused Coordination Trust in Partner's Contribution	12 11	23 (88.46%)	Face-to-face Comfort Non-verbal Cues	10 9	19 (73.08%)
	Negative	Communication Lag	3	3 (11.54%)	Over-awareness of Partner Initial Shyness	2 5	7 (26.92%)
Technology Role	Positive	Joint Focus Reduced Real-world Distractions	10 10	20 (76.92%)	Supported Mutual Awareness Enabled Non-verbal Coordination	13 9	22 (84.62%)
	Negative	Limited Non-verbal Cues	6	6 (23.08%)	Distractions from Physical Environment	4	4 (15.38%)

6 DISCUSSION

6.1 Addressing RQ1: Differences in Social Presence and Collaborative Engagement Between AR and VR Conditions

Our findings suggest that AR and VR both support social presence and collaborative engagement through different pathways. In general, the item-level analysis of the social presence module (SP-5, SP-6) verified increased mutual attention, aligning with the methods showing that the use of interactive games can influence each other [42]. In VR conditions, participants experience intense immersion and deep flow states, while those in AR condition exhibit increased behavioral engagement and improved sensitivity to social cues, particularly significant in short-term interactions between unfamiliar teammates.

Participants in VR condition reported significantly deeper sensory immersion and stronger flow states than those in AR. They frequently characterized the experience as one of complete absorption within the game environment, accompanied by a decreased awareness of time passage. For instance, Participant 8B stated:

"My teammates and I were completely immersed in the puzzle in the VR environment, not paying attention to how long it took me to solve the puzzle."

The increased challenge and imaginative involvement observed in participants in VR conditions promote sustained focus and exploration in a collaborative setting [44], aligned with flow theory. These results corroborate existing studies that indicate that VR induces stronger presence and absorption than AR [51, 17], although such immersion may also attenuate immediate social awareness [52, 48].

In contrast, participants in the AR condition demonstrated significantly stronger behavioral involvement. They reported continuously monitoring their partner's actions and adapting their own behaviors in real time. Participant 3A noted:

"I can always see what my teammates are doing in the game and help him when he makes a mistake."

AR maintains real-world cues such as gaze, gestures, and body orientation, facilitating synchronous collaboration among unfamiliar teammates. This study also showed that AR significantly improves situational awareness and joint regulation in collaborative tasks with physical objects, outperforming VR in this aspect, as noted in [33, 43].

AR also amplified emotional resonance, although it simultaneously intensified negative affective experiences ($p < 0.001^{***}$, $d = 1.148$). Participant 13A reported:

"I noticed that when I made a mistake, my teammate expressed his frustration through subtle gestures, even though he did not say anything, which made me blame myself."

In line with normative social influence theory [34], these findings suggest that AR experiences amplify concerns about social evaluation, reflecting increased social sensitivity in AR condition collaborations.

6.2 Addressing RQ2: Different Challenges and Limitations in Social Interactions for Collaborative Games Between AR and VR Conditions

Participants identified different challenges and limitations that affected their social interactions in each environment. The advantages of collaboration in VR condition were significant with increased immersion, focused attention, and exploratory engagement, while its primary limitations involved increased physiological burden and interpersonal synchronization challenges. In contrast, collaboration in AR condition benefited from increased sensitivity to teammate behaviors and cooperation within a shared physical space, but its constraints included emotional contagion and the attentional costs of divided focus.

Intense task focus in VR can also momentarily reduce awareness of one's teammate, potentially reducing social perception.

Quantitative results reinforced the specific constraints of VR. Although participants in VR condition showed significantly higher sensory immersion and flow, along with increased perceived challenge and positive post-game experience, they also experienced significantly higher nausea and oculomotor discomfort. Existing studies indicate that cybersickness is multifactorial, modulated by immersion or presence, and visually induced motion cues in addition to locomotion, so differences between VR and AR can contribute to the discomfort patterns observed between conditions [54, 5]. Intense focus in the VR condition can momentarily reduce awareness of the social teammate. This aligns with the feedback from P10 who stated:

"I think VR is great for deep thinking and solo experimentation, but when I work together, I need to make an extra effort to stay in sync with my teammate."

This indicates that maintaining group dynamics in highly immersive collaborative task settings requires clear alignment strategies [24]. Practically, co-located VR collaboration mechanisms may benefit from explicit alignment mechanisms (e.g., shared task-state cues) to help unfamiliar pairs remain synchronized. Practically, co-located VR collaboration mechanisms may benefit from explicit alignment mechanisms (e.g., shared task-state cues) to help unfamiliar pairs remain synchronized.

Meanwhile, participants in the AR condition exhibited significantly greater behavioral involvement and mutual attention than those in the VR condition, but also reported higher levels of negative affect. This suggests that increased social sensitivity can amplify individual social discomfort, as indicated by interview responses. Participant 13A stated:

“My teammate’s subtle facial and bodily expressions influenced my judgment, significantly increasing my self-consciousness.”

This finding highlights that increased visibility of teammate gestures intensifies social concerns and emotional contagion, aligning with the normative social influence theory, which describes anxiety arising from perceived evaluation [34]. In these settings, when teammates’ reactions are visible, subtle frustration cues can be perceived as evaluative, causing participants to hesitate, over-monitor their behavior, or blame themselves after errors. Furthermore, these findings are supported by evidence that dual-channel processing of physical and virtual information in AR condition increases the demands of divided attention [45]. Even though body gestures can improve coordination [39] and co-located visibility reduces communication costs [25] between teammates under AR condition, they also experienced cognitive load when constant distractions from teammates occur during task execution [7], producing a double-edged effect. For AR, designs can mitigate split attention and evaluative pressure by simplifying attention switching between physical navigation and virtual task cues.

6.3 Summary of Key Findings

The findings illustrate complementary pathways to collaboration for AR and VR conditions. VR facilitates collaborative engagement through deep immersion and task absorption, while AR promotes collaboration through improved behavioral coordination and mutual attention. For unfamiliar pairs, both conditions provide effective but distinct routes to a shared goal, involving a fundamental trade-off between deep immersion and social awareness.

The challenges identified by the participants were condition-specific. The primary limitations of VR involved physiological burdens and the increased effort required for interpersonal synchronization, which can reduce social awareness. In contrast, in the AR condition, the high visibility of teammates’ reactions heightened users’ self-consciousness, and, coupled with the need to split attention between real-world navigation and virtual task cues, increased negative feelings and attentional load during coordination. This situation suggests a central trade-off where VR prioritizes individual immersion at a potential cost to social coordination, while AR promotes mutual regulation at a potential cost to increased cognitive and emotional load.

6.4 Limitations and Future Work

The sample comprised primarily university students (undergraduate and postgraduate), which limits generalizability to other cohorts.

This study presents several limitations. The sample consisted predominantly of university students, limiting the generalizability of the findings to different cohorts. Given our focus on familiarity with VR and AR environments, this study restricted recruitment to students; including non-students or broader geographic diversity would have introduced additional confounding and was beyond the

scope of this study. In addition, the study design confounded the locomotion demands: the AR condition required room-scale movement, whereas the VR condition was largely stationary. Although this manipulation improved ecological realism for AR, it compromised internal validity by making it difficult to isolate condition-specific effects. Consequently, observed differences may be due to navigation demands and divided attention rather than to the condition itself. This constraint may also have influenced discomfort outcomes, and future work will evaluate VR condition in a larger, obstacle-free tracked space to better match locomotion between conditions. Finally, the study relied heavily on self-rated ratings and did not include objective behavioral or performance measures.

Future work should explore additional types of asymmetry, beyond informational, positional and visual, to evaluate their effects on immersion, coordination, communication depth, and cognitive load under AR and VR conditions. The study should also consider collaborative scenarios outside the escape room paradigm, such as serious games designed for professional training, to include participants from diverse cultural and occupational backgrounds. In addition, future studies should include more in-depth evaluation metrics, such as in-game performance data, analysis of gestures and communication patterns, or physiological signals (such as EEG or fNIRS), and conduct experiments with larger sample sizes to more fully explain the strengths and limitations of AR and VR, thus strengthening design guidelines for technology-condition collaboration.

7 CONCLUSION

This study presents an analysis to explore the differences in collaborative engagement and social presence in VR and AR during asymmetric puzzle-solving game tasks among unfamiliar users. The analysis reveals clear process-level differences: VR promotes greater immersion, flow and exploratory engagement, albeit with a higher physiological burden; by contrast, AR increases mutual attention and coordination, but can increase self-consciousness and evaluative pressure when teammates’ reactions are highly visible. Despite these differences, overall collaborative satisfaction remains comparable under conditions, suggesting that environmental effects shape social engagement in different ways, each with its own merits and limitations. By mapping each environment’s unique characteristics to its respective advantages and drawbacks, this study highlights the complementary roles of VR and AR in collaborative task design, with the objective of improving collaboration among unfamiliar users.

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